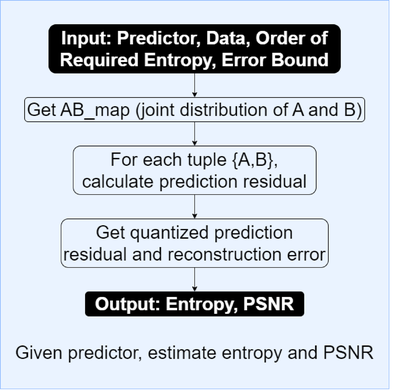
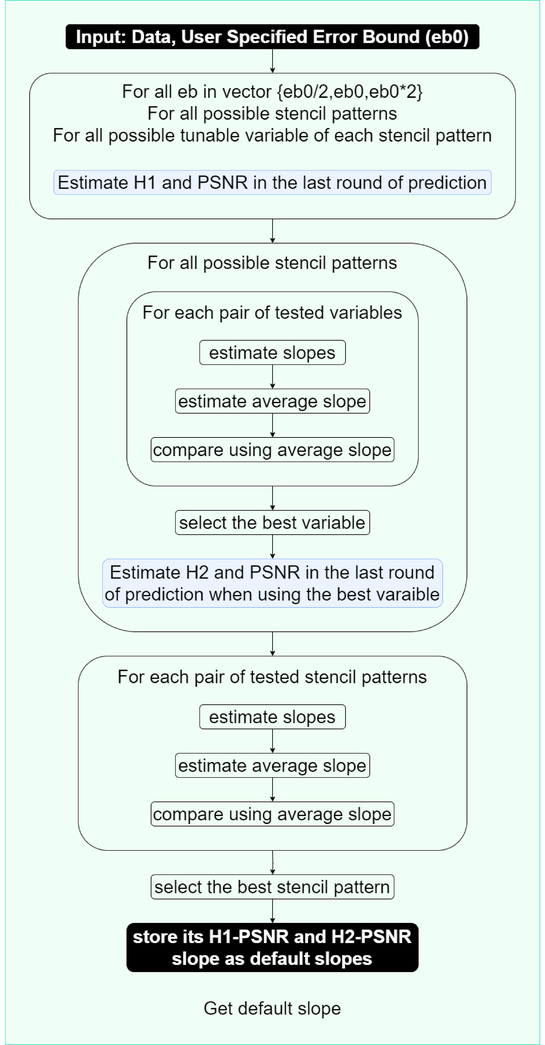
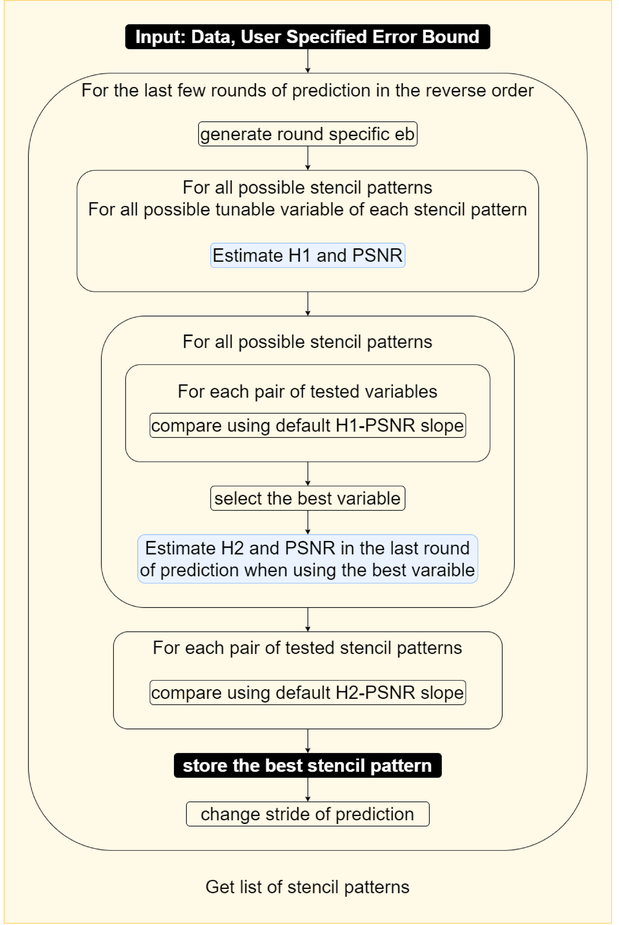
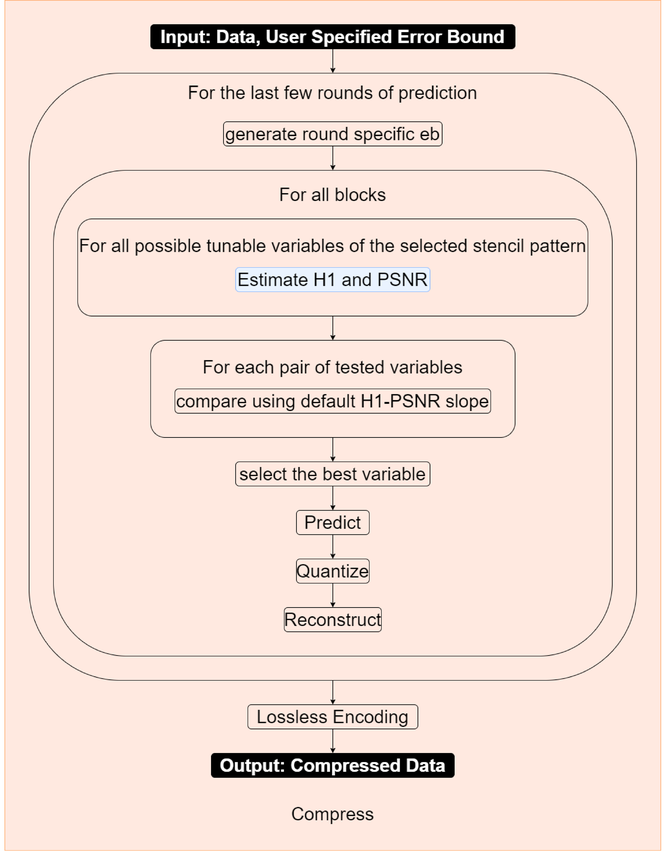
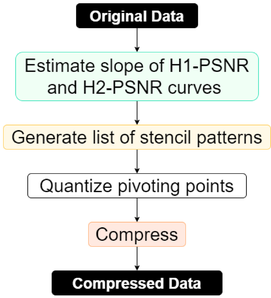

Input: Predictor, Data, Order of Required Entropy, Error Bound
Get AB_map (joint distribution of A and B)
For each tuple {A,B}, calculate prediction residual
Get quantized prediction residual and reconstruction error
Output: Entropy, PSNR
Given predictor, estimate entropy and PSNR


Input: Data, User Specified Error Bound (eb0)
For all eb in vector {eb0/2,eb0,eb0*2}
For all possible stencil patterns
For all possible tunable variable of each stencil pattern
Estimate H1 and PSNR in the last round of prediction
For all possible stencil patterns
For each pair of tested variables
estimate slopes
estimate average slope
compare using average slope
select the best variable
Estimate H2 and PSNR in the last round of prediction when using the best varaible
For each pair of tested stencil patterns
estimate slopes
estimate average slope
compare using average slope
select the best stencil pattern
store its H1-PSNR and H2-PSNR slope as default slopes
Get default slope


Input: Data, User Specified Error Bound
For the last few rounds of prediction in the reverse order
generate round specific eb
For all possible stencil patterns
For all possible tunable variable of each stencil pattern
Estimate H1 and PSNR
For all possible stencil patterns
For each pair of tested variables
compare using default H1-PSNR slope
select the best variable
Estimate H2 and PSNR in the last round of prediction when using the best varaible
For each pair of tested stencil patterns
compare using default H2-PSNR slope
store the best stencil pattern
change stride of prediction
Get list of stencil patterns


Input: Data, User Specified Error Bound
For the last few rounds of prediction
generate round specific eb
For all blocks
For all possible tunable variables of the selected stencil pattern
Estimate H1 and PSNR
For each pair of tested variables
compare using default H1-PSNR slope
select the best variable
Predict
Quantize
Reconstruct
Lossless Encoding
Output: Compressed Data
Compress


Original Data
Estimate slope of H1-PSNR and H2-PSNR curves
Generate list of stencil patterns
Quantize pivoting points
Compress
Compressed Data